Practice Set A

Class 12 Mathematics, Chapter 7: Integrals

Maximum marks: **54** **20** questions

Name _____ Date _____ Score _____ / 54

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1 — 8 marks

1 MCQ, PYQ-style — 1 mark

$\int e^x \sec x\,(1+\tan x)\,dx$ is equal to:

(A) $e^x \cos x + C$ **(B)** $e^x \sec x + C$ **(C)** $e^x \sin x + C$ **(D)** $e^x \tan x + C$

2 MCQ, CBSE SQP 2025-26 — 1 mark

If $\int x^3 \sin^4(x^4)\cos(x^4)\,dx = a\sin^5(x^4) + C$, then a is equal to:

(A) $-\frac{1}{10}$ **(B)** $\frac{1}{20}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{1}{5}$

3 MCQ, PYQ-style — 1 mark

The value of $\int_0^{\pi/2} x\sin x\,dx$ is:

(A) 0 **(B)** 1 **(C)** $\frac{\pi}{2}$ **(D)** $\frac{\pi}{2} - 1$

4 MCQ, CBSE SQP 2024-25 — 1 mark

$\int \frac{dx}{x^3\,(1+x^4)^{1/2}}$ equals:

(A) $-\frac{1}{2x^2}\sqrt{1+x^4} + C$ **(B)** $\frac{1}{2x}\sqrt{1+x^4} + C$

(C) $-\frac{1}{4x}\sqrt{1+x^4} + C$ **(D)** $\frac{1}{4x^2}\sqrt{1+x^4} + C$

5 MCQ, CBSE SQP 2025-26 — 1 mark

If $f(a+b-x) = f(x)$, then $\int_a^b x\,f(x)\,dx$ is equal to:

(A) $\frac{a+b}{2}\int_a^b f(b-x)\,dx$ **(B)** $\frac{a+b}{2}\int_a^b f(a-x)\,dx$

(C) $\frac{b-a}{2}\int_a^b f(x)\,dx$ **(D)** $\frac{a+b}{2}\int_a^b f(x)\,dx$

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** $\int_{-1}^{1} \log\left(\frac{2-x}{2+x}\right) dx = 0$. **Reason (R):** If f is an even function, then $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, PYQ-style Find $\int \frac{\sin x - \cos x}{\sqrt{1+\sin 2x}}\,dx,\ 0 < x < \frac{\pi}{2}$.	2 marks

Level 2

19 marks

8	Short answer, PYQ-style Evaluate: $\int_{0}^{\pi/4} \tan^2 x\,dx$.	2 marks
9	Short answer, CBSE SQP 2023-24 Find: $\int \sqrt{\frac{x}{1-x^3}}\,dx;\ x \in (0,1)$.	2 marks
10	Short answer, CBSE SQP 2023-24 Find: $\int \frac{2x^2+3}{x^2(x^2+9)}\,dx;\ x \neq 0$.	3 marks
11	Short answer, PYQ-style Evaluate: $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$.	3 marks
12	Short answer, CBSE SQP 2024-25 Evaluate: $\int \left\{\frac{1}{\log x} - \frac{1}{(\log x)^2}\right\} dx;\ x > 1$.	3 marks
13	Short answer, CBSE SQP 2025-26 Sketch the graph of $y = \lvert x+1 \rvert$. Evaluate $\int_{-4}^{2} \lvert x+1 \rvert\,dx$. What does the value of this integral represent on the graph?	3 marks
14	Short answer, CBSE 2025 Evaluate: $\int_{0}^{\pi/4} \frac{dx}{\cos^3 x\sqrt{2\sin 2x}}$.	3 marks

Q No.	Question	Marks
Level 3		**27 marks**
15	Long answer, PYQ-style Evaluate: $\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x}\, dx.$	5 marks
16	Long answer, CBSE SQP 2025-26 Find: $\int \frac{(3\sin\theta - 2)\cos\theta}{5 - \cos^2\theta - 4\sin\theta}\, d\theta.$	5 marks
17	Long answer, CBSE SQP 2025-26 Evaluate: $\int_0^{1} \frac{\log(1+x)}{1+x^2}\, dx.$	5 marks
18	Short answer, PYQ-style Find: $\int \frac{x+3}{\sqrt{5-4x-x^2}}\, dx.$	3 marks
19	Long answer, PYQ-style Evaluate: $\int_0^{\pi/4} \frac{\sin x + \cos x}{9 + 16\sin 2x}\, dx.$	5 marks
20	Case study, Board-pattern **Case Study:** While revising definite integrals, a student notes the property $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$. When the lower limit is 0, it becomes $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$. Use these properties to answer the following.	4 marks
	(i) Evaluate $\int_2^8 \frac{\sqrt{10-x}}{\sqrt{x}+\sqrt{10-x}}\, dx.$	1 mark
	(ii) Evaluate $\int_0^{\pi/2} \log(\tan x)\, dx.$	1 mark
	(iii) Evaluate $\int_0^{\pi/2} (2\log\sin x - \log\sin 2x)\, dx.$	2 marks

Answer key

Chapter 7: Integrals, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) $e^x \sec x + C$
2	(B) $\dfrac{1}{20}$
3	(B) 1
4	(A) $-\dfrac{1}{2x^2}\sqrt{1+x^4} + C$
5	(D) $\dfrac{a+b}{2}\displaystyle\int_a^b f(x)\,dx$
6	(B) Both true; R is not the correct explanation of A
7	$-\log\lvert\sin x + \cos x\rvert + C$

Level 2

8	$1 - \dfrac{\pi}{4}$
9	$\dfrac{2}{3}\sin^{-1}\left(x^{3/2}\right) + C$
10	$-\dfrac{1}{3x} + \dfrac{5}{9}\tan^{-1}\dfrac{x}{3} + C$
11	$\dfrac{\pi}{12}$
12	$\dfrac{x}{\log x} + C$
13	9; it is the area bounded by $y = \lvert x+1\rvert$, the x-axis and the lines $x = -4$, $x = 2$.
14	$\dfrac{6}{5}$

Level 3

15	$\dfrac{\pi(\pi-2)}{2}$
16	$3\log(2-\sin\theta) + \dfrac{4}{2-\sin\theta} + C$
17	$\dfrac{\pi}{8}\log 2$
18	$-\sqrt{5-4x-x^2} + \sin^{-1}\left(\dfrac{x+2}{3}\right) + C$
19	$\dfrac{1}{40}\log 9$
20	(i) 3 (ii) 0 (iii) $-\dfrac{\pi}{2}\log 2$